

30 Supergraphs

The introduction of Feynman graphs in the late 1940s provided the enormous advantage of preserving manifest Lorentz invariance at every stage of perturbative calculations. For this purpose it was necessary that the exchange of all spin states of any virtual particle should be described by a single propagator. Fortunately, it has become possible to go a step further, and develop a supergraph formalism, in which supersymmetry as well as Lorentz invariance is kept manifest at every stage [1]. To do so, it is necessary to describe the exchange of all the particles described by a given superfield by a single superpropagator.

There is a problem here in that the left-chiral superfields Φ over which we integrate are subject to a differential constraint, $\mathcal{D}_R\Phi = 0$. This is analogous to the problem in electrodynamics that the field-strength tensor is subject to a differential constraint, the homogeneous Maxwell equation. In electrodynamics this problem is dealt with by expressing the field-strength tensor in terms of a vector potential, and doing path integrals over the vector potential instead of the field-strength tensor. In much the same way, here we will impose the left-chiral constraint by expressing left-chiral superfields in terms of potential superfields, over which we integrate in path integrals. In this formalism we encounter problems like those produced by gauge invariance in electrodynamics, and deal with them in much the same way.

The most important result to come out of the supergraph formalism was the non-renormalization theorem for the superpotential [2]. We have already proved this theorem in Section 27.6, using a much easier indirect technique developed by Seiberg, which can be extended also to describe non-perturbative effects. Nevertheless, it is interesting to see how these surprising cancellations of renormalization effects occur in actual perturbative calculations.

30.1 Potential Superfields

Consider a theory of left-chiral superfields $\Phi_n(x, \theta)$ and their complex conjugates, but for simplicity without gauge superfields. All of the vacuum expectation values of time-ordered products of component fields may be calculated from vacuum expectation values of the time-ordered products of these superfields. We might try to calculate these from the path-integral formula

$$\begin{aligned} & \langle T \{ \Phi_{n_1}(x_1, \theta_1), \Phi_{n_2}(x_2, \theta_2), \dots \} \rangle \\ &= \int \left[\prod_{n, x, \theta} d\Phi_n(x, \theta) \right] \exp(iI[\Phi]) \Phi_{n_1}(x_1, \theta_1) \Phi_{n_2}(x_2, \theta_2) \cdots \end{aligned} \quad (30.1.1)$$

where $I[\Phi]$ is the action

$$I[\Phi] = \frac{1}{2} \int d^4x \left[\sum_n \Phi_n^*(x, \theta) \Phi_n(x, \theta) \right]_D + 2 \operatorname{Re} \int d^4x [f(\Phi)]_{\mathcal{F}}. \quad (30.1.2)$$

(The factor $1/2$ is introduced in the first term, as in Eq. (26.4.3), to give the component fields of Φ a conventional normalization.) But we cannot simply read off the supergraph

Feynman rules from Eq. (30.1.1), because the functional integral over the superfields Φ_n must be constrained to satisfy the left-chiral condition $\mathcal{D}_R \Phi_n = 0$.

This is analogous to a similar problem in electrodynamics. As discussed in Section 12.3, at energies below the electron mass the interactions of soft photons with each other are described by an effective action of the form:

$$I[f] = -\frac{1}{4} \int d^4x \left[f_{\mu\nu} f^{\mu\nu} + c_1 (f_{\mu\nu} f^{\mu\nu})^2 + c_2 (\epsilon_{\mu\nu\rho\sigma} f^{\mu\nu} f^{\rho\sigma})^2 \right].$$

But we cannot read off the Feynman rules from this formula without taking into account the fact that the path integral is constrained by the homogeneous Maxwell equations

$$\partial_\mu f_{\nu\rho} + \partial_\nu f_{\rho\mu} + \partial_\rho f_{\mu\nu} = 0.$$

As everyone knows, we deal with this constraint by introducing a four-vector potential A_μ , with $f_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, so that the constraint is automatically satisfied, and integrate over $A_\mu(x)$, not $f_{\mu\nu}(x)$.

In the same way, we may adopt a trick already used in Section 26.6 to derive the field equations for superfields, and introduce a non-chiral potential superfield $S_n(x, \theta)$, with

$$\Phi_n = \mathcal{D}_R^2 S_n, \quad (30.1.3)$$

where

$$\mathcal{D}_R^2 \equiv \sum_{\alpha\beta} \epsilon_{\alpha\beta} \mathcal{D}_{R\alpha} \mathcal{D}_{R\beta}, \quad (30.1.4)$$

so that Φ_n automatically satisfies the left-chiral constraint, $\mathcal{D}_{R\alpha} \Phi_n = 0$. In place of Eq. (30.1.1), we then have the path-integral formula

$$\begin{aligned} & \langle T \{ \Phi_{n_1}(x_1, \theta_1), \Phi_{n_2}(x_2, \theta_2), \dots \} \rangle \\ &= \int \left[\prod_{n,x,\theta} dS_n(x, \theta) \right] \exp(iI[\mathcal{D}_R^2 S]) \mathcal{D}_R^2 S_{n_1}(x_1, \theta_1) \mathcal{D}_R^2 S_{n_2}(x_2, \theta_2) \cdots \end{aligned} \quad (30.1.5)$$

In expressing the action (30.1.2) in terms of S_n , we may recall that the D -term of a superderivative does not contribute to the action, so we may shift the operator $(\mathcal{D}_R^2)^* = \mathcal{D}_L^2$ acting on S_n^* in the first term of Eq. (30.1.2) to act instead on S_n , which gives

$$I[\mathcal{D}_R^2 S] = \frac{1}{2} \int d^4x \left[\sum_n S_n^* \mathcal{D}_L^2 \mathcal{D}_R^2 S_n \right]_D + 2 \operatorname{Re} \int d^4x [f(\mathcal{D}_R^2 S)]_{\mathcal{F}}. \quad (30.1.6)$$

The $\mathcal{D}_R^2$ operator in any one of the factors $\mathcal{D}_R^2 S$ in any term in $f(\mathcal{D}_R^2 S)$ may be taken to act on the whole term, since $\mathcal{D}_R$ gives zero when acting on any other factor of $\mathcal{D}_R^2 S$ in that term. In this way, we may write

$$f(\mathcal{D}_R^2 S) = \mathcal{D}_R^2 \tilde{f}(S), \quad (30.1.7)$$

where $\tilde{f}(S)$ is obtained from $f(\mathcal{D}_R^2 S)$ by omitting any one operator $\mathcal{D}_R^2$ in each term. For instance, for a single type of superfield, if $f(\Phi) = \sum_r c_r \Phi^r$, then

$$\tilde{f}(S) = \sum_r c_r S (\mathcal{D}_R^2 S)^{r-1}.$$

Using Eqs. (30.1.6), (30.1.7), and (26.3.31), we may put the whole action in the form of a D -term

$$I[\mathcal{D}_R^2 S] = \frac{1}{2} \int d^4 x \sum_n [S_n^* \mathcal{D}_L^2 \mathcal{D}_R^2 S_n]_D + 2 \operatorname{Re} \int d^4 x [\tilde{f}(S)]_D. \quad (30.1.8)$$

Eq. (26.6.5) shows that this may also be written as a superspace integral:

$$I[\mathcal{D}_R^2 S] = -\frac{1}{4} \int d^4 x \int d^4 \theta \sum_n S_n^* \mathcal{D}_L^2 \mathcal{D}_R^2 S_n - \operatorname{Re} \int d^4 x \int d^4 \theta \tilde{f}(S). \quad (30.1.9)$$

This is the action we will use in the path-integral derivation of the supergraph formalism.